

Procedure 3-1

Setting up management access using the console or USB port

Use this procedure to connect to Waveserver Ai via the console or USB-C port and acquire or provision an IPv4 or IPv6 address and gateway for the management network interface. This procedure also describes the steps to connect Waveserver Ai to the management network and log in.

Note: The DCN and ILAN interfaces described in this procedure are intended for intra-building use only. In customer premises locations where compliance to Telcordia GR-1089-CORE intra-building lightning is required, install shielded twisted pair (STP) wire with shield terminations crimped into the RJ-45 connectors for connection at both ends.

Prerequisites

- If you are assigning a static IPv6 address and gateway, the terminal supports IPv6
- One of the following cables is available:
 - A USB-C-to-USB-C cable or a USB-C-to-USB-A cable (dependent on compatibility with your PC or laptop)
 - A DB9-to-RJ45 or USB-to-RJ45 (straight connection) console cable

Note: External adapters are not required except when the Waveserver Ai is connected to a complete non-standard proprietary console port.

- An Ethernet cable is available.
- The management network supports IPv4 or IPv4 + IPv6.
- Waveserver Ai line ports are not physically connected to photonic system equipment, and transceivers are not installed in Waveserver Ai client ports.



CAUTION

Risk of laser radiation exposure

Do not look into the end of optical fiber connectors or into faceplate connectors. The light source used in fiber-optic devices can damage your eyes. Optical/laser operation should be verified by observing status LEDs. Do not look into the laser to verify operation.

- A terminal emulator (for example, PuTTY) is open on the terminal.

Procedure 3-1 (continued)

Setting up management access using the console or USB port

- If you are using a Macintosh computer or MacBook to connect to Waveserver Ai, Ciena recommends installing Minicom for serial port setup. Minicom is available at the Web site:
<http://macappstore.org/minicom/>
- If you are using a MacBook 2016 or later, you must use a USB-C-to-USB-A dongle to access the Waveserver Ai console port. The cable must have the following configuration: MacBook USB-C (female) port -> USB-C (male) - USB-A (female) cable attached to a USB-A (male) to USB-C (male) cable -> Waveserver Ai USB-C (female) port.

Step	Action								
1	Connect Waveserver Ai to the management network: <table><tr><th>If</th><th>Then</th></tr><tr><td>You are setting up a group of interconnected Waveserver Ai nodes</td><td>Connect the first Waveserver Ai node in the group to the management network (router). Use an Ethernet cable to connect the Waveserver Ai DCN port to the router. Then, use an Ethernet cable to connect the ILAN 2 port on the Waveserver Ai to the ILAN 1 port on the next Waveserver Ai node in the group. See Option 1 - ILAN interconnection for DCN access.</td></tr><tr><td>You are connecting a Waveserver Ai directly to the management network</td><td>Use an Ethernet cable to connect the DCN port on the Waveserver Ai to the management network (router). See Option 2 - Direct connection to DCN.</td></tr></table> Note: <i>Group</i> refers to a set of interconnected Waveserver Ai units that are connected to the same photonic system.	If	Then	You are setting up a group of interconnected Waveserver Ai nodes	Connect the first Waveserver Ai node in the group to the management network (router). Use an Ethernet cable to connect the Waveserver Ai DCN port to the router. Then, use an Ethernet cable to connect the ILAN 2 port on the Waveserver Ai to the ILAN 1 port on the next Waveserver Ai node in the group. See Option 1 - ILAN interconnection for DCN access .	You are connecting a Waveserver Ai directly to the management network	Use an Ethernet cable to connect the DCN port on the Waveserver Ai to the management network (router). See Option 2 - Direct connection to DCN .		
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2	Connect the terminal to Waveserver Ai. <table><tr><th>If you are</th><th>Then go to</th></tr><tr><td>connecting the terminal (PC or laptop) to the Waveserver Ai USB-C port using a USB-C-to-USB-C or USB-C-to-USB-A cable</td><td>step 3</td></tr><tr><td>connecting the terminal (Macintosh or MacBook) to the Waveserver Ai USB-C port using a USB-C-to-USB-A cable</td><td>step 6</td></tr><tr><td>connecting the terminal to the Waveserver Ai console port using a console port cable</td><td>step 13</td></tr></table>	If you are	Then go to	connecting the terminal (PC or laptop) to the Waveserver Ai USB-C port using a USB-C-to-USB-C or USB-C-to-USB-A cable	step 3	connecting the terminal (Macintosh or MacBook) to the Waveserver Ai USB-C port using a USB-C-to-USB-A cable	step 6	connecting the terminal to the Waveserver Ai console port using a console port cable	step 13
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Procedure 3-1 (continued)

Setting up management access using the console or USB port

- 3 Connect a USB-C cable to the Waveserver Ai USB-C port and then to the PC or laptop.
The PC detects the cable and automatically installs any required drivers.
Note: If the drivers do not automatically install, download and install CP210x USB to UART Bridge Virtual COM Port (VCP) from the website:
<https://www.silabs.com/products/development-tools/software/usb-to-uart-bridge-vcp-drivers>
- 4 Open the Device Manager on your PC or laptop and under the Ports (COM & LPT) section, determine the COM port used to connect to Waveserver Ai.
- 5 With the terminal emulator, configure the connected terminal with the following settings:
 - Serial line: COM port noted in [step 4](#)
 - Port: a supported baud rate (the USB-C port supports baud rates up to 921600)
 - Connection type: Serial
 Go to [step 15](#).
- 6 Connect a USB-C-to-USB-A cable to the Waveserver Ai USB-C port and then to the Macintosh computer or MacBook.
The Macintosh computer or MacBook detects the cable and automatically installs any required drivers.
Note: If the drivers do not automatically install, download and install CP210x USB to UART Bridge Virtual COM Port (VCP) for Macintosh from the website:
<https://www.silabs.com/products/development-tools/software/usb-to-uart-bridge-vcp-drivers>
- 7 Restart the Macintosh computer or MacBook.
- 8 Verify the presence of the USB cable by searching the `/dev/cu` folder.
- 9 Open a terminal window and connect to Waveserver Ai. Ciena recommends using Minicom for the connection.
- 10 If you are using Minicom, in the terminal window, type `minicom -s`.
- 11 Use the arrow keys to scroll to `Serial port setup` and press Enter to select it.
- 12 Configure the connected terminal with the following settings;
 - Serial Device: USB device noted in [step 8](#)
 - Bps/Par/Bits: 9600 8N1
 - Hardware Flow Control: No
 - Software Flow Control: No
 Go to [step 15](#).

Procedure 3-1 (continued)

Setting up management access using the console or USB port

- 13 Connect a console port cable to the Waveserver Ai console port and then to the PC or laptop.
- 14 With the terminal emulator, configure the connected terminal with the following RS-232 settings:
 - Speed: Default value
 - Data bits: 8
 - Stop bits: 1
 - Parity: None
 - Flow control: None
- 15 At the prompt, press the Enter key for auto-baud rate detection.
A login prompt appears with the correct baud rate displayed.
Note 1: The RJ-45 port supports baud rates up to 115200.
Note 2: The baud rate can be changed if you terminate the login session using the logout command, or if the login session automatically terminates due to inactivity. You can also change the baud rate by disconnecting the cable, changing the baud rate on the terminal, and then reconnecting the cable (USB or RJ-45).
Note 3: The auto-baud prompt is displayed every 30 seconds or whenever a key is pressed. Auto-baud detection supports these rates: 9600, 14400, 19200, 38400, 57600, 115200, 230400, 460800, and 921600.
- 16 At the login prompt, enter a username and a password to access the Waveserver Ai. Log in as `su`.

Procedure 3-1 (continued)

Setting up management access using the console or USB port

17	Note the following:	
If	Then	
The Waveserver Ai DHCPv4 client successfully acquires an IP address and gateway from a DHCPv4 service on the management network	With the terminal emulator, retrieve the provisioned IP address:	
	<code>interface show</code>	
	Go to step 19	
The DHCPv4 client cannot acquire an IP address and gateway from a DHCPv4 service on the management network	Check for and resolve issues with the DHCPv4 services on the management network, and reattempt to acquire a DHCPv4 IP address and gateway.	
Waveserver Ai successfully acquires an IPv6 SLAAC IP address and gateway from the management network	With the terminal emulator, retrieve the provisioned IP address:	
	<code>interface show</code>	
	Go to step 19	
Waveserver Ai cannot acquire an IPv6 SLAAC IP address and gateway from the management network	Check for and resolve issues with the router configuration, and reattempt to acquire an IPv6 SLAAC IP address and gateway.	
You are assigning a static IPv4 IP address and the management network that Waveserver Ai will be connected to might support DHCPv4 services, which will automatically assign an IPv4 IP address that will overwrite the static IP address you are provisioning	With the terminal emulator, disable the DHCPv4 client on Waveserver Ai:	
	<code>dhcp client disable</code>	
You are assigning a static IPv6 IP address and the management network that Waveserver Ai will be connected to might support SLAAC, which will automatically assign an IPv6 IP address that will overwrite the static IP address you are provisioning	With the terminal emulator, prevent Waveserver Ai from accepting an auto-assigned SLAAC IP address:	
	<code>interface ipv6-stack set accept-router-advertisement off</code>	

Procedure 3-1 (continued)

Setting up management access using the console or USB port

- 18 Assign the IP address and gateway:

Type	Use the commands
IPv4	<code>interface set interface local ip <IP address / mask> [gateway <IP address>]</code>
	<code>interface set gateway <IP address></code>
IPv6	<code>interface add interface local ip <IP address / mask></code>
	<code>interface add gateway <IPv6 address></code>

- 19 With the terminal emulator, log in to Waveserver Ai via an SSH connection using the provisioned IP address. Enter the user name (`su`) of the default user account and a short sequence of any characters for the password.

Note 1: Although there is no password associated with the default super user account, you must enter a short sequence of any characters as the password to complete the login.

Note 2: Ciena strongly recommends that you set a password for the default user account as soon as possible after you log in to Waveserver Ai for the first time. The alarm “Credentials Warning” will activate to remind you to do this. For more information, see [User accounts](#) and [Credentials Warning](#).

- 20 Disable unused communications interfaces on Waveserver Ai (see figures [Option 1 - ILAN interconnection for DCN access](#) and [Option 2 - Direct connection to DCN](#)):

If	Then
The Waveserver Ai is the first in a group (i.e., DCN port is connected to the management network),	Disable the ILAN 1 port: <code>chassis port disable port ilan-1</code>
The Waveserver Ai is being added to an existing group with access to the management network	Disable the DCN port: <code>chassis port disable port dcn</code>
The Waveserver Ai is being added to an existing group with access to the management network and no other Waveserver Ai will be connected to it	Disable the DCN port and the ILAN-2 port: <code>chassis port disable port dcn</code> <code>chassis port disable port ilan-2</code>

Procedure 3-1 (continued)

Setting up management access using the console or USB port

- 21** Save the provisioned settings to the configuration file:
- ```
configuration save
```
- 22** Do one of the following:
- Go to [Manually setting the system clock](#)
  - Go to [Configuring NTP MD5 authentication](#) and [Configuring the NTP client](#).
  - Go to [User authentication](#).
  - Go to [User accounts](#).
  - Go to [Commissioning Waveserver Ai](#).
  - Close the connection to Waveserver Ai.

—end—